

Agricultural Extension RAG

Smart Retrieval for Farmers

Team Gambia | TRI AI Saturdays Cohort 10

Final result: $nDCG@5 = 0.870$ (private leaderboard) • 8th place overall

The Problem

Retrieval quality caps the quality of any downstream RAG answer

Vocabulary mismatch

Farmers say "matooke wilt"; factsheets say "Banana Bacterial Wilt / *Xanthomonas campestris*"

Hard negatives

Documents built to look similar but answer a different problem — e.g. potassium vs. nitrogen deficiency

Graded, multi-doc relevance

~13.6 judged documents per query, on a 0–3 relevance scale — ranking order matters, not just recall

The Dataset

695 agricultural extension factsheets, 308 labeled training queries

695

documents

308

train queries

4,194

relevance labels

200

test queries

12 crops covered • Sources: FAO, CGIAR, IITA, ICRISAT, Plantwise, AGRA, national extension services

Relevance scale: 3=perfect • 2=relevant • 1=marginal (same issue, wrong crop) • 0=hard negative

Our Approach

Three complementary retrieval strategies, combined

1

Lexical (BM25/TF-IDF)

Exact term matching, stemmed

2

Dense (bi-encoder)

Semantic similarity via embeddings

3

Cross-encoder rerank

Joint query+doc attention on top candidates

Every combination and parameter (fusion weights, stemming vs. lemmatization, reranker choice) was A/B tested against real labeled data before being adopted — not assumed from architecture alone.

Key Finding: Vocabulary Normalization

Traced to a real, diagnosed failure case

“How can I prevent bean anthracnose?”

The correct document’s title: “Preventing Bean Anthracnose...” — but “prevent” ≠ “preventing” as exact tokens.

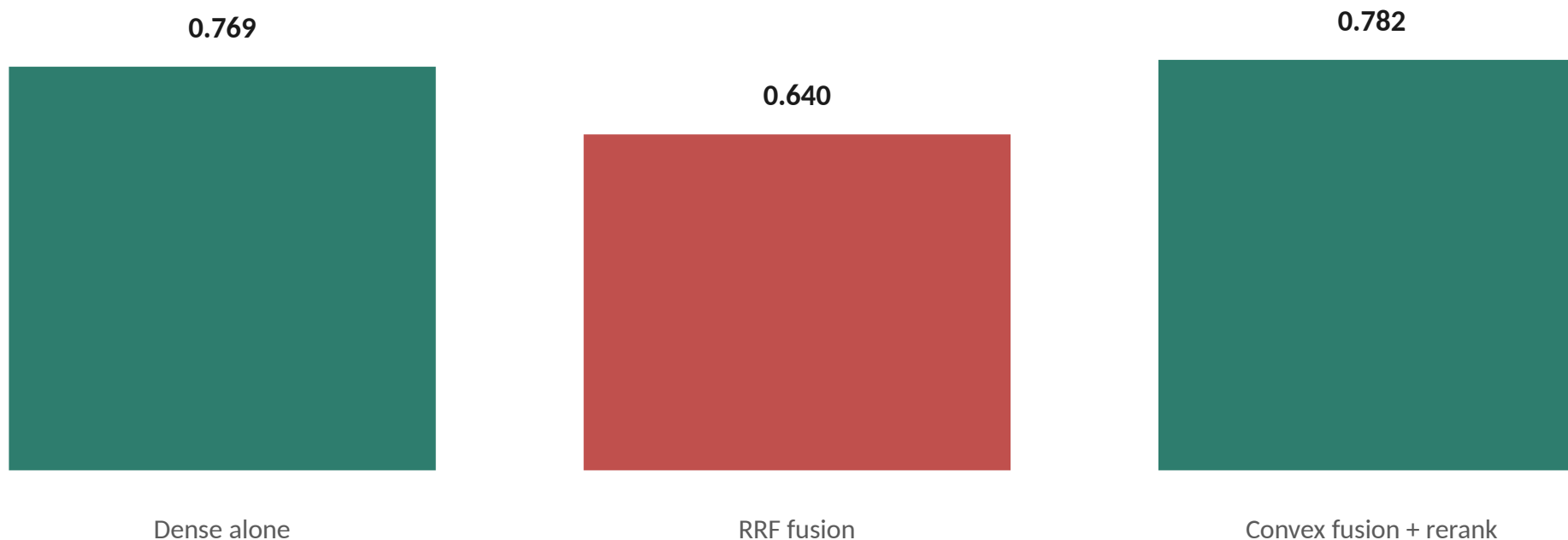
Method	Before stemming	After stemming	Gain
TF-IDF	0.494	0.582	+0.088
BM25	0.469	0.512	+0.044

Largest single validated improvement in the entire pipeline.

Key Finding: How We Combine Retrievers

Convex combination beats Reciprocal Rank Fusion (RRF)

RRF only sees rank position — it can't express "trust dense much more than BM25 here." Measured directly: RRF actively hurt.



Key Finding: Bigger Isn't Always Better

A larger reranker underperformed, at ~12x the compute cost

	ms-marco-MiniLM-L-6-v2	bge-reranker-base
Parameters	22M	278M
Local nDCG@5	0.782 (winner)	0.747
Relative compute	1x	~12x

Model capacity alone doesn't guarantee better results on a narrow domain the larger model was never fine-tuned for.

Results

Validated with 5-fold cross-validation, not a single number

Method	nDCG@5 (local)
TF-IDF (stemmed)	0.582
BM25 (stemmed, tuned)	0.512
Dense (bge-small)	0.769
Convex fusion	0.770
Fusion + MiniLM rerank	0.782

5-fold CV: 0.782 ± 0.024 • Final leaderboard: Public 0.838, Private 0.870 • 8th place overall

Limitations & Impact Considerations

- Residual retrieval errors could feed wrong context into a downstream generator — this is a retrieval-only evaluation, not an end-to-end safety validation
- Uneven document distribution across crops (128 Maize docs vs. 10 Pearl millet) likely means uneven retrieval quality across crops
- Recurring failure pattern found via error analysis: near-identical documents differing only by phrasing (“manage X” vs. “outbreak of X”) sometimes retrieve very differently for functionally identical questions
- Stakeholder engagement was dataset-mediated, not field-validated with real farmers or extension officers — flagged as an open limitation, not hidden

Thank You

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[github.com/\[org\]/C10-team-gambia](https://github.com/[org]/C10-team-gambia)